MIL-STD-1873A(AT)
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MASS FINISHING, PROCESS FOR

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FOREWORD

This Military standard briefly describes mass finishing processes and their applications.

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1. SCOPE

1.1 Purpose. This standard describes procedures, equipment types, and precautions to be considered in selection of methods for mass finishing metallic and non-metallic parts.

1.2 Scope. This standard covers conventional mass finishing processes currently in use.

1.3 Classification. Basic established mass finishing techniques include the following:

a. Barrel (rotational) finishing
b. Vibratory finishing
c. Centrifugal (high pressure) finishing
    1. Disc
    2. Barrel
d. Spindle finishing

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2. REFERENCE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

OCCUPATIONAL SAFETY & HEALTH ADMINISTRATION (OSHA)

(Application for copies should be addressed to US Department of Labor, Occupational Safety & Health Administration, Washington, D.C. 20210.)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR METALS (ASM)

METALS HANDBOOK

EIGHTH EDITION	- Volume 2, "Heat Treating, Cleaning and Finishing" - Mass Finishing Definitions.
NINTH EDITION	- Volume 5, "Surface Cleaning, Finishing and Coating" - Mass Finishing.

(Copies of the above publications may be obtained from the American Society for Metals, Metals Park, OH 44073.)

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3. DEFINITIONS

3.1 Definitions. Refer to ASM, Volume 2, Eighth Edition for definitions not listed.

3.2 Barrel finishing. Barrel (rotational) finishing involves the tumbling or rolling of parts in rotating barrel type containers. Finishing action occurs primarily by the sliding movement in the upper layer of the workload as the barrel rotates.

3.3 Vibratory finishing. Vibratory finishing involves vibrating the workload at a predetermined frequency to cause a constant movement of the entire mass.

3.4 Centrifugal barrel finishing. Centrifugal (high pressure) barrel finishing combines spinning of the work container about its axis for agitation along with the high speed counter-rotation of the turret or shell mounting the work container to produce high pressure rubbing contact.

3.5 Centrifugal disc finishing. Centrifugal disc finishing involves high speed spinning of a cylinder bottom disc. This action forces the mass outward and then upward against the stationary cylinder wall resulting in fast cutting or finishing action.

3.6 Spindle finishing. Spindle finishing involves fixturing or chucking the work piece to a spindle(s) and immersing the part(s) into a rotating or vibratory tub containing the abrasive media.

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4. GENERAL REQUIREMENTS

4.1 Equipment.

4.1.1 Barrel machines, conventional types. Barrel shapes, sizes and materials used in the construction, including the need for liners, are dependent upon the shape, size, and type of material being processed. Unlined steel or cast iron barrels are typically used for processing to clean (or remove heat treat scale from) ferrous castings and forgings. Typical types used in industry are (see figure 1):

a. Open-ended, tilting type.
b. Bottle-neck barrel adaption.
c. Horizontal barrel.
d. Submerged barrel.

4.1.2 Vibratory machines. Vibratory machines are basically open-topped of either a tub or bowl configurations (see figure 2 & 3).

4.1.3 Centrifugal machines.

4.1.3.1 Disc process. Disc finishing machines are constructed of open-topped, heavy walled, stationary vertical cylinders. The bottoms, as shown in figure 4, are formed from discs designed to spin at high speeds, yet retain the work load and media.

4.1.3.2 Barrel process. Barrel finishing machines are comprised of containers (drums) mounted on the periphery of a turret. As the turret rotates at a high speed in one direction, the drums spin at a slower speed in the opposite direction (see figure 5).

4.1.3.3 Spindle process. Spindle finishing machines consist of a circular rotating container, which holds the abrasive media, and a rotating or oscillating spindle to which the part is affixed.

4.2 Auxiliary equipment.

a. Separating equipment
b. Loading equipment
c. Temperature controlled water source
d. Drainage facilities
e. Waste treatment facilities

4.3 Materials.

4.3.1 Finishing compounds. Finishing compounds present the physical and chemical agents which act on the material to be finished. They may be either water or supensions in water. They are designed primarily for one-time use. Compounds are classified according to the purpose of the operation either as non-abrasive or abrasive, each including several types.

4.3.1.1 Non-abrasive compounds. Non-abrasive compounds based on pH (degree of acidity or alkalinity) have specific functions on the material of the part being processed.

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4.3.1.1.1 Cleaning compounds. The following non-abrasive compounds are used for cleaning purposes.

a. Acid (pH 1-6) types
b. Alkaline (pH 8-14) types
c. Emulsion, emulsifiable solvents

4.3.1.1.2 Burnishing compounds. Burnishing compounds must establish and maintain proper solution pH to prevent corrosion of the parts while the compound removes the dirt and oxides from the parts being finished. Burnishing compounds are usually alkaline (pH 9-12) for this reason.

4.3.1.2 Abrasive compounds. Abrasive compounds consist of one or more types, and grit sizes of abrasive grains compounded with various cleaners. Most common types are aluminum oxide and silicon carbide.

4.3.1.2.1 Grinding compounds. Grinding compounds should contain abrasives and synthetic nonlubricating detergents. The compound should keep the medium active and the work pieces clean so that abrading action is not impaired.

4.3.1.3 Other compounds. Certain mass finishing processes require both non-abrasive and abrasive compounds.

4.3.1.3.1 Rustproofing compounds. Rust preventives are usually chemical compounds or oils. They are used in both non-abrasive and abrasive compounds whenever ferrous parts are to be mass finished.

4.3.1.3.2 Deburring compounds. Deburring compounds may contain non-abrasive or abrasive materials depending on the medium employed and the degree of deburring that is required.

4.3.1.4 Compound functions. The finishing compound has the following functions:

1. To develop and maintain cleanliness of parts and media during the process.
2. To control pH, foam, and water hardness.
3. To wet surfaces.
4. To emulsify oily soils.
5. To remove tarnish or scale.
6. To control part color.
7. To suspend soils.
8. To control lubricity.
9. To prevent corrosion.
10. To provide cooling.
11. To ensure effluent, meeting Environmental Protection Agency (EPA), Office of Safety and Health Administration (OSHA), and plant standards.

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4.3.1.5 Compound introduction. Additions of compounds (liquid or powder) to the mass finishing equipment may be made by:

1. Batch
2. Recirculation
3. Flow-through

4.3.2 Finishing media. Media is material added to the mix to abrade or burnish edges and surfaces of components to be finished and to keep parts separate from one another to avoid or limit part-on-part impingement. Although a variety of products are used, see 4.3.2.1 through 4.3.2.6, the media field is dominated by the use of aluminum oxide and silicon carbide.

4.3.2.1 Synthetic random media. Fused and sintered aluminum-oxide media, crushed and graded, are available in a number of grades, both heavy cutting and fine finishing. Generally they are much tougher than natural media and more consistant (see 4.3.2.5).

4.3.2.2 Preformed ceramic media. Porcelain or other vitreous material are mixed with abrasives and formed into shapes, then fired to vitrify. These media are available in a large range of shapes and sizes. This type of material is the present standard for the mass finishing industry.

4.3.2.3 Preformed resin-bonded media. These are abrasive bonded into polyester or urea-formaldehyde resins. Like the ceramic materials, these are available in a broad range of shapes and sizes with different types, grades, and quantities of abrasive to meet a range of applications. Plastic media are somewhat softer than ceramic and for a given degree of abrasion, usually have shorter life. These softer materials achieve better preplate finish than other media and are better suited for handling soft metals.

4.3.2.4 Steel media. Hardened steel preformed shapes are available in a variety of shapes and sizes and are well suited for burnishing. Steel applications also include cleaning and light deburring.

4.3.2.5 Natural media. Stones that have been quarried, crushed, and graded were the original media for mass finishing operations. Natural media have been largely replaced by synthetic materials which are harder with longer life, greater consistency of cut, wear, and dimension, and greater variety of capabilities.

4.3.2.6 Agricultural media. Saw dust, corncob, and walnut shells are frequently used in mass finishing machines for drying. Mixed with fine abrasives, these materials are suited for some fine polishing operations. Wood pegs may be coated with fine abrasives or waxes for edge radiusing and finishing of some wooden and plastic components.

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5. DETAIL REQUIREMENTS

5.1 Selection of process. Factors to be considered when selecting the most suitable mass finishing process shall include:

a. Production requirements

1. Size and shape (configuration) of parts.
2. Variety of parts (small, large, mixed).
3. Part material and hardness.
4. Batch size.
5. Adaptability of process to automation.

b. Quality considerations.

1. Quality of parts to be processed.
2. Quality of parts expected after processing.
3. Surface finish.
4. Uniformity of part-to-part.
5. Uniformity over edges and surfaces.
6. Cleanliness.
7. Edge condition.

5.2 Selection of media. Factors affecting the final selection of the best media usually shall be based upon experience or trial and error:

a. Shape and size of media

1. To remove burrs.
2. To achieve uniform edge and surface finishes.
3. To avoid jamming in holes and recesses.
4. To achieve ease of separation.
5. To achieve shortest cycle time.

b. Availability and cost of media

1. From consistent supplier.
2. For economical cost per unit weight or volume.
3. For consistent quality.

c. Ability and versatility of media

1. For minimum wear and reclassification
2. For handling a range of products within a given machine.
3. For minimum break in requirements.
4. For cushioning action between parts.

5.3 Barrel operations.

5.3.1 Cleaning. Oil, grease, dirt, scale, and machining chips must be removed from work before it is subjected to barrel finishing processes, such as deburring or burnishing. Cleaning can be accomplished by conventional immersion or spray techniques, or in a barrel, as a preliminary operation.

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5.3.2 Pre-inspection. Quality inspections shall be done on parts for nicks, dents, scratches etc. before the work pieces undergo mass finishing operations.

5.3.3 Deburring. Deburring by mass finishing can be accomplished on stampings, machined parts, die castings, small sand castings, forgings, and sintered metal parts.

5.3.4 Burnishing. Burnishing is employed when work pieces need very smooth surfaces. Burnishing removes a small amount of surface metal and developes a high luster.

5.3.5 Post-inspection. Quality inspections shall be done on work pieces after mass finishing operations.

5.4 Process instructions and control. Process instructions should cover all variables. The variables include:

a. Machine cycle, including process times, barrel speeds, or vibratory frequency and amplitude, or gravitational (g) force and speed, depending on equipment type.
b. Load levels, total load, and proportion of parts to media.
c. Compound and water flow rates.
d. Media and compound variables.
e. Pretreatment and post-treatment variables.

5.5 Advantages and disadvantages. Advantages and disadvantages of the various mass finishing processes are detailed in table I.

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6. NOTES

6.1 Safety precautions. Recommended safety precautions in mass finishing include:

a. Closed barrel locks which do not release covers until internal gas pressure build up has been relieved.
b. Ventilation of the barrel, preferably automatic.
c. Precaution against rotating machinery.
d. Use of rubber gloves and face masks while handling acids and alkalis.

6.2 Causes of difficulties in mass finishing. Causes of difficulties commonly encountered in mass finishing are:

a. Excessive impingement on part surface.

1. Insufficient amount of media in chamber.
2. Parts in work load are too large for the machine.
3. Excessive equipment speed or amplitude.
4. Low solution level or flow rate.
5. Insufficient or wrong compounds.
6. Undesirable handling methods.
7. Media too large for part size.
8. Wrong type of media.

b. Roll over of edges, corners, and burrs.

1. Abrasive action too slow.
2. Wrong type of media.
3. Media particles too large for part size.
4. Excessive equipment speed or amplitude.
5. Excessive work load for volume of media.
6. Incorrect water level.
7. Processing time too long.

c. Persistent loading of media in holes, slots, or recessed areas.

1. Wrong size of media.
2. Wrong shape of media.
3. Excessive wear or depreciation of media.
4. Media fracturing during operation.
5. Failure to select media as required.

d. Poor surface finish.

1. Insufficient reduction in process.
2. Inadequate flushing and cleaning of equipment.
3. Incorrect compound.
4. In-process corrosion.
5. Water flow not correct.
6. Media too aggressive or too hard.
7. Part-on-part impingement.

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6.3 Waste disposal. Some compounds used in mass finishing are harmful and their disposal may pose problems. In all instances, Federal and local regulations should be checked before establishing any procedure for waste disposal.

6.4 Supersession data. This standard supersedes MIL-STD-1873(AT), 24 January 1983.

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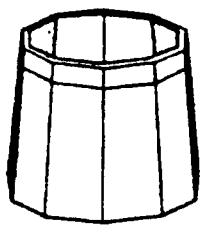
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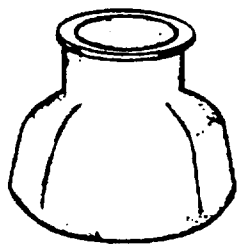
TABLE I. Advantages and disadvantages of mass finishing processes.

Process	Advantage	Disadvantage
Barrel	Low initial cost. Low operating cost for supplies. Very low equipment maintenance cost.	Slow process. Skillful operator essential. Automation impractical. Wet working area. No in-process inspection.
Vibratory:		
Tub	Fast operation. Handles all part sizes. Open for in-process inspection. Practical full batch automation. Practical in-line automation.	Slower than high energy process. External material handling required.
Bowl	Open for in-process inspection. Practical full batch and continuous operations. Requires no auxiliary equipment to automate. Internal separation. Space saving. Lowest cost for general purpose work. Simple selection and operation with little operator skill required. Can produce cleanest parts and excellent surface finsih.	Somewhat slower than vibratory tub.
Centrifugal:		
Disc	Fast processing. Open for in-process inspection. Practical automated batch process. Compact operation.	Part size limitation. High initial investment.
Barrel	Fast processing. Fragile part handling. High precision part handling. Potential automated batch processing. Automatic change from grind to super finish. Produces finest finish. Improves fatigue strength.	No in-process inspection. High initial investment.
Spindle	Fast processing. Possible automation, robot load and unload No part-on-part impingement.	Limited part geometry. Parts must be fixtured. High labor cost.

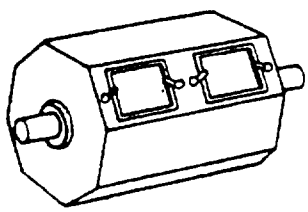
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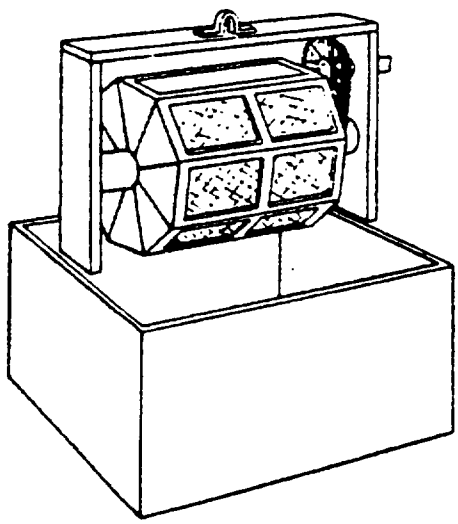
Open-ended, tilting type



Bottlenecked type



Horizontal type



Submerged type

FIGURE 1. Barrels, conventional types.

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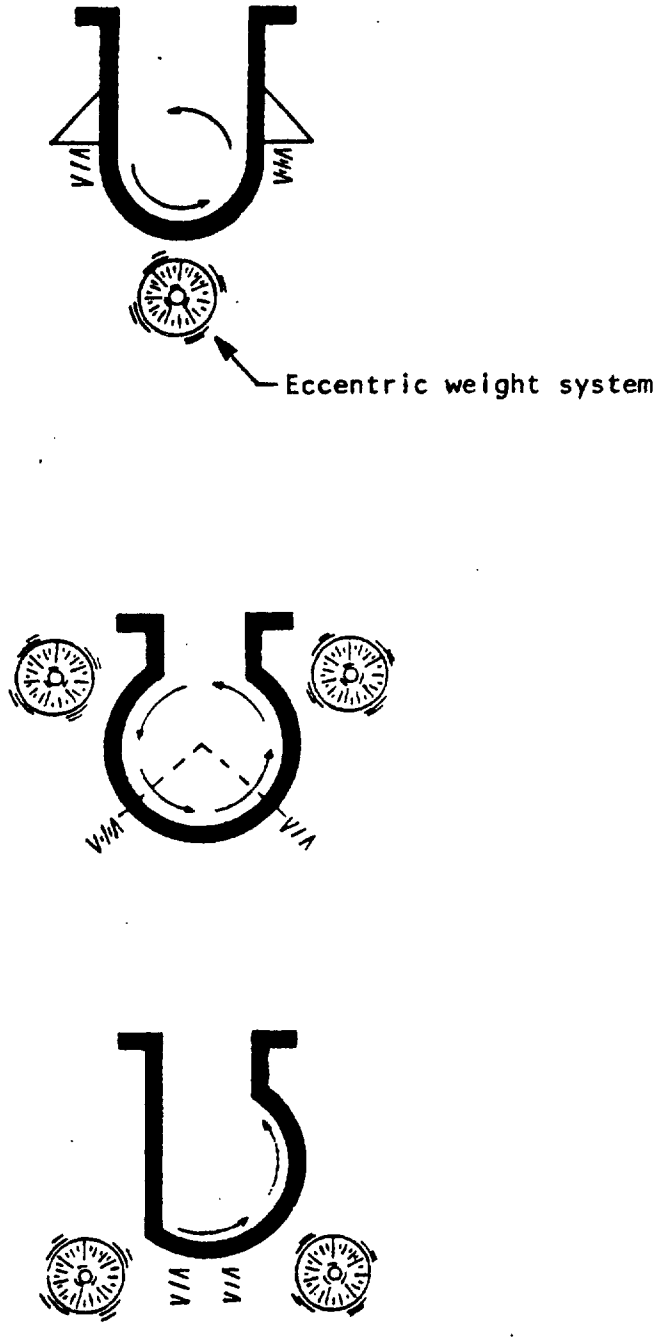
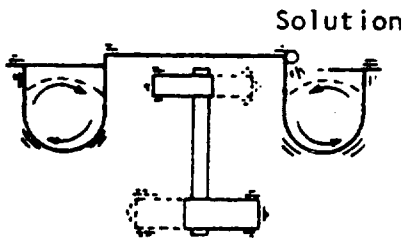
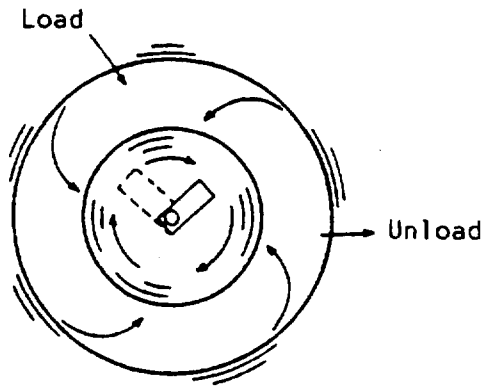


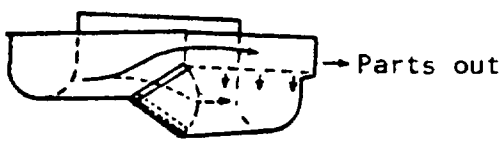
FIGURE 2. Tub vibrators.

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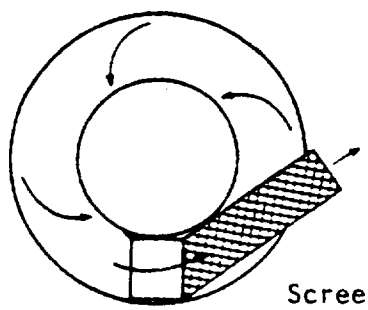


Eccentric weight system,
flat-bottom style





Deflector or gate raised,
non-flat-bottom style



Screen separator

FIGURE 3. Bowl vibrators.

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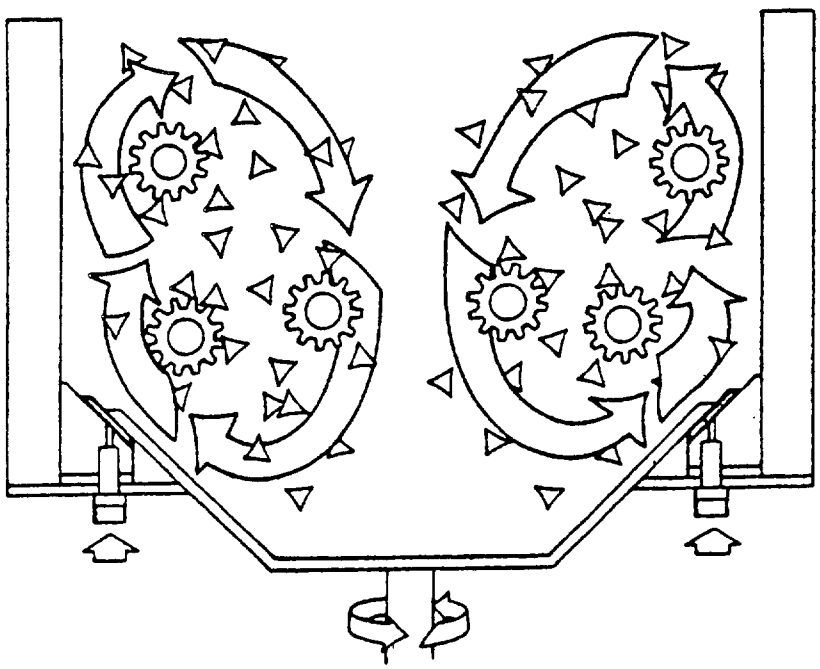
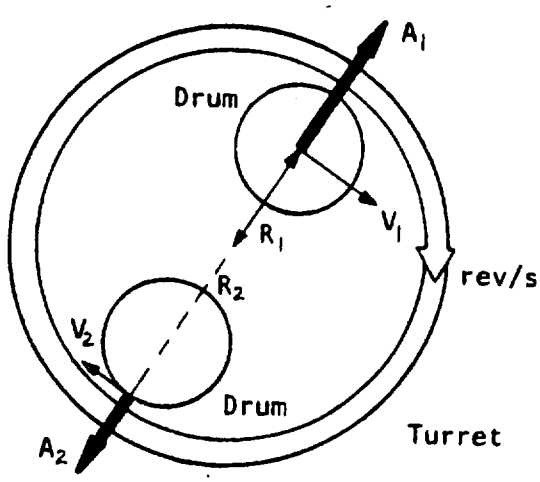
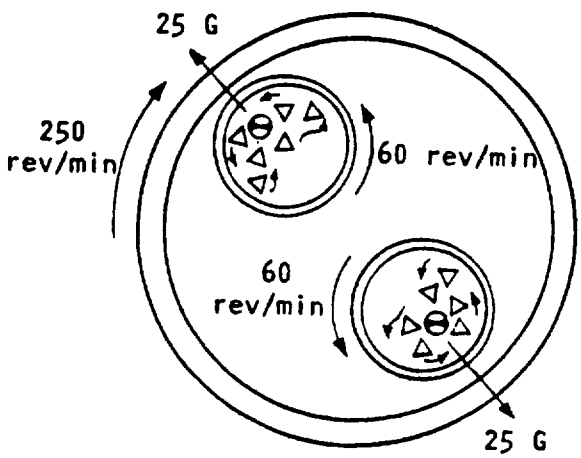


FIGURE 4. Centrifugal disc machine.

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Principle of centrifugal barrel process



Action within a centrifugal barrel machine

FIGURE 5. Centrifugal barrel machine.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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